The last step is to take the inner product with $<A|$. We do that by expanding the bra $<A|$ in eigenvectors on the right-hand side, and then using the orthonormality of the eigenvectors. The result is

$$\langle A|\mathbf{L}|A\rangle = \sum_i (\alpha_i^* \alpha_i)\lambda_i. \qquad (4.13)$$

Using the probability principle (Principle 4) to identify $(\alpha^*_i \alpha_i)$ with the probability $P(\lambda_i)$, we immediately see that the expression on the right side of Eq. 4.13 is the same as the expression on the right side of Eq. 4.11. That is to say,

$$\langle \mathbf{L} \rangle = \langle A|\mathbf{L}|A\rangle. \qquad (4.14)$$

Thus, we have a quick rule to compute averages. Just sandwich the observable between the bra and ket representations of the state-vector.

In the previous lecture (Section 3.5), we promised to explain how the action of a Hermitian operator on a state-vector is related to the results of physical measurements. Armed with our knowledge of expectation values, we can now keep that promise. If we look back at Eq. 3.21, we see an example of an operator, σ_z, acting on state-vector $|r>$ to produce a new state-vector. We can view this equation as half of the calculation for the expectation value of the measurement σ_z—the right-hand part of the sandwich, if you will. The rest of that calculation involves taking the inner product of this state-vector with the dual vector $<r|$.So when σ_z acts on $|r>$ in Eq. 3.21, it produces a state-vector from which we can calculate the probabilities of each σ_z measurement outcome.

4.8 IGNORING THE PHASE-FACTOR

In previous lectures, we said that we can ignore the overall phase-factor of a state-vector, and promised to explain why in a later section. Having worked out the rule for averages, we'll take a short detour to keep that promise.